

Development of mobile application - Computação Móvel 24/25 - Parallel Connect

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CONTRIBUTIONS

For this project, we both agreed we worked the same amount, meaning the contribution is 50/50. We both worked on various features, meaning we both used API calls and firebase so we could have a large understanding of all the project and not just one part, so it was well distributed.

REPOSITORY

https://github.com/rosabarbara/Parallel_Connect.git

CONCEPT

Our mobile application is called Parallel Connect, which is an app that lets users connect to spirits in the "afterlife". While walking in the world, you can find lost spirits and talk to them. You can understand and try to find out more about their story. It also allows the users to interact with objects the spirits may have lost in some places and capture memories for the future. It is an app that focuses on connecting this life with the "other" letting you interact in various ways with life after death.

RELATED WORK

At the beginning of the development process, we considered several existing apps and shows that could relate to our concept. These sources inspiration for functionalities and features that aligned with our vision, such as:

- **Whatsapp**
Its an app that allows people to communicate with other through messages, video calls, and voice calls. Our app extends this by allowing users not only to interact with other users through objects "left by spirits," but also to engage in direct conversations with spirits themselves, providing a better social experience.
- **Pokemon Go**
A location-based game where the user needs to move around to catch Pokemons. This inspired our feature of finding and interacting with spirits and objects in the real world, creating a dynamic exploration experience where users can discover new elements around them.
- **Stranger Things**
The show's portrayal of connecting to another dimension through technology and communication inspired the concept of bridging the gap between worlds.

LIBRARIES/SENSORS

Libraries

- **Google Maps API**
Interactive map to see where we are and spirit and objects locations.
- **Firebase Realtime Database**
Store user information aswell as chat messages and everything related to storage.
- **CameraX API**
Used to take photos and save them
- **Gemini API**
Used to make spirits talk - gave them a prompt "You are this: respond to this " plus the message the user sent

Sensors

- **Location**
Used to get the current user location for spirit tracking and map integration.
- **Camera**
Used to captures photos of moments important to the user for later reference in the gallery.

FEATURES COMPLETED

- **Authentication System**
We implemented a secure registration and login system, ensuring personalized experiences for users. Firebase Authentication was used to manage account creation and login functionality.
- **Interactive Map**
We developed a real-time map interface where users can locate and interact with spirits. Custom markers were implemented to represent spirits and objects on the map, dynamically updating based on the user's location.
- **Spirit Conversations**
We implemented interactive chat screens where users can initiate conversations with spirits. Each spirit has a unique chat interface, with messages stored in the Firebase Realtime Database to ensure consistency across devices.
- **Forum Interaction**
We developed a forum feature where users can discuss objects left by spirits, such as "Jeff's missing shoes." Posts and messages are updated in real-time using Firebase.
- **Photo Capture and Gallery**
We created a photo capture feature that allows users to save moments related to spirit interactions in their personal gallery. Photos are securely stored in Firebase Storage and referenced in the Firebase Realtime Database for easy access.
- **Profile and Progress Tracking**
We added a profile screen that displays user details, such as

their name and the number of spirits they've caught. This feature highlights individual user achievements.

PROCESS

First Idea

In our first pitch we decided the context of the app and how we we're thinking about some of the main features. We decided right away that we wanted to use the GPS to walk around the world and find spirits so we could interact with them. We also wanted to have some ways of connecting users with each other. So, first, we thought about creating some sort of messages between users but, as that did not make sense for our application, we decided to changed the idea and came up with this sort of conversation forum where you can tal with the users if you find a certain object. We also thought about a feature - "events" - where an astrological event would happen and something could change in our app while it was on. After this first brainstorming process, the main idea was clear and we could start the development.

Development

After creating some low-fidelity prototypes, we started coding our application. Before going into detail and worrying about some API's like the gemini, we created the menus and the Map. From there, we created screens for everything we had in mind and then we started implementing the API's, sensors and also the connection to the database. While some things were easy like the Gemini API, other things were much more difficult to implemente, like the Magnetometer, that we wanted to use to check levels of energy in a certain location, so we decided to scratch that idea off.

Some ideas came up also while we were developing the app already, such as the use of the camera to register some important moments we had within the application, like when finding a spirit. The idea was to share those moments with the other users, but, unfortunately, we ended up not having enough time to do that so the user is only allowed to see its own photos.

Testing

The final stage of this project was testing the application. We wanted to check that every single feature created was both working well and ready for the demo. More importantly, we wanted to see if the firebase was also working well with and for different users. While everything was working fine for what we had, we also found some last minute minor bugs, for example, we had a problem with the way we were showing the images on the forum because of the bitmap size and we, as couldn't find a solution, we had to take the image off. Because of that the object screen ended up not being as intuitive as we initially wanted it to be.

WHAT WE ACCOMPLISHED

- **Build an application that could only be a Mobile App**
Our app leverages mobile-specific features like GPS tracking and camera integration, which are not feasible in a non-mobile environment.
- **Leverage the Device Sensors to provide features that make sense for your app**

We used sensors like the GPS and the camera to track user's movement and let it interact with spirits/objects and capture key moments.

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